GAUTIER DAGAN

Machine Learning Engineer, M.Sc. in AI

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EDUCATION

M.Sc. in Artificial Intelligence

University of Amsterdam

September 2017 – August 2019

- Graduated Cum Laude
- GPA: 8.6/10
- Relevant Coursework:
  - Machine Learning I, II
  - Natural Language Processing, Computer Vision
  - Deep Learning, Deep Learning for Natural Language Technology

B.A. in Computer Science, Economics, and International Studies (Triple Major)

Northwestern University

September 2013 – June 2017

- GPA: 3.6/4.0
- Honour Roll (Dean's List): W 2016, S 2016, F 2016, S 2017
- Recipient of the International Student Scholarship and the UWC Davis Scholars Scholarship

EXPERIENCE

Machine Learning Engineer

Mansa

September 2019 – Present

First Employee at Mansa - a fintech startup seeking to revolutionise lending for freelancers

- Created a credit score to analyse and estimate risk based on financial health indicators
- Communicated algorithmic design and findings to corporate lenders
- Developed and architected the data pipeline from scratch using AWS, Docker and FastAPI
- Completed the AWS Cloud Practitioner Certification

Graduate Student Researcher

Institute of Logic, Language and Computation

January 2019 – September 2019

Member of the i-machine-think research group at the ILLC

- Supervised peers and guided research into design of analysis methods for emergent languages
- Submitted papers to major NLP conferences

Teaching Assistant

University of Amsterdam

February 2019 – April 2019

Teaching Assistant in Information Retrieval course for the M.Sc. in Information Studies (Data Science track)

- Covered topics such as distributed crawling, Term/Semantic Search and Learning to Rank
- Supervised students to create a search engine from scratch

Co-founder

Zcruit

February 2016 – January 2018

Co-founded Zcruit – a predictive analytics and data-aggregation start-up focusing on college football recruiting

- Conceptualised and brought to market a Minimum Viable Product, now sold nationwide to 40+ enterprise clients
- Developed an AWS based scalable and serverless backend architecture with Lambda, DynamoDB, SQS
- Designed an asynchronous scraper able to refresh >50,000 individual data-points daily and minimized cloud infrastructure costs

PROJECTS

M.Sc. Research Thesis

Research on language evolution, bridging traditional Linguistic theories and Deep Learning

- Simulated language evolution through Referential Games and Neural Architecture Search algorithms
- Analysed emerged languages in a multi-agent and multi-modal setting
- Available as Arvix Preprint

Location Attention for Extrapolation to Longer Sequences

Independent research project through the ILLC to replicate and expand on Location Attention

- Experimented with new attention mechanisms and tested on Language Modelling baselines
- Implemented state-of-the-art Sequence-to-Sequence models and outperformed baseline metrics using proposed technique
- Submitted to ACL 2020, available as Arvix Preprint

ECCV 2018 Workshop Paper: Adding Object Detection Skills to Visual Dialogue Agents

Published research paper at the 2018 EECV Workshop on Shortcomings in Vision and Language

- Adapted a Mask R-CNN in a Visual Dialog agent to improve performance on the GuessWhat?! task
- Replaced ground truth images using predictions allowing, for the first time, inference on unlabelled images

Global and Local Critical Policy Learning for Abstractive Summarization

Research project for the 2018 Information Retrieval II course

- Replicated and surpassed state-of-the-art abstract summarisation methods by optimising ROUGE score directly instead of through standard Maximum-Likelihood-Estimation

SKILLS

Programming:

Python | Pytorch | JavaScript | NodeJS | C++

Other Technologies:

Git | AWS | Latex | Adobe Suite | Microsoft Office

Languages:

English (native), French (native), Spanish (A2)

Competitions:

Kaggle Competition Expert (**top 2%**), RoboCup IranOpen 2018, Autonomous Robot Design Competition (2016 and 2017)